Data and Preprocessing

Dataset Types: Structured vs. Unstructured

- **Structured Data:** Highly organized and formatted for easy analysis. Typically stored in tables (like SQL databases) with rows and columns. Examples: customer databases, transaction logs. Easy to search, query, and analyze due to a fixed schema.
- **Unstructured Data:** Lacks a predefined structure. Cannot be stored logically in tables/rows/columns. Includes text documents, emails, images, audio, video, and social media content. Requires special tools (e.g., NoSQL databases, text analysis software) for analysis.

Why it's important: The type dictates storage, processing method, and suitable analytical tools.

Features and Labels

- **Features:** Also called independent variables, attributes, or predictors. These are measurable properties or characteristics of the data used as input for prediction. Can be numerical, categorical, or text-based. Example: For house price prediction—number of rooms, area, location.
- **Labels:** The target variable or outcome to be predicted/classified. Also called dependent variable. In supervised learning, labels are known during training. Example: House price, disease status (yes/no), or email label (spam/ham).

Importance: Features supply the "evidence"; labels are the "answers" the model learns to predict.

Feature Engineering

- **Definition:** The process of transforming raw data into features that better represent the underlying problem to the predictive models, thus improving their performance. Involves creating, selecting, transforming, or combining features.
- **Importance:**
    - Improves prediction accuracy.
    - Reduces overfitting.
    - Enhances model interpretability.
    - Makes model training more efficient.

Feature Scaling: Normalization vs. Standardization

Feature Scaling brings all features onto a comparable scale to ensure fair treatment by algorithms (especially those measuring distances, like k-NN, SVM, K-means).

- **Normalization (Min-Max Scaling):** Scales data to a given range, usually.

  Formula: $X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$

  Use when features have different scales and do not have outliers.
- **Standardization (Z-Score Normalization):** Rescales features so they have zero mean and unit variance.

  Formula: $X_{std} = \frac{X - \mu}{\sigma}$

  Use when data approximates a normal distribution or contains outliers.

	Normalization	Standardization
Scale range	or [-1,1]	Not bounded
Method	Min & max values	Mean & standard deviation
Outlier sensitivity	Sensitive	Robust
When to use	No outliers, various scales	Normal dist. or outliers present

Dimensionality Reduction

- **Definition:** The process of reducing the number of features (dimensions) in a dataset while preserving as much relevant information as possible.
- **Common Techniques:**
    - Feature Selection: Choosing a subset of important features (e.g., filter, wrapper, embedded methods).
    - Feature Extraction: Transforming features into a lower-dimensional space (e.g., PCA, t-SNE, UMAP, Autoencoders).
- Importance:
    - Reduces computational cost.
    - Limits overfitting.
    - Improves visualization and interpretability.

Data Splitting: Train/Test/Validation

- Training Set: Used to fit the model.
- Validation Set: Used to tune parameters and prevent overfitting (often used in model selection and hyperparameter tuning).
- Test Set: Used to evaluate the final model performance on unseen data.

Why it’s important: Prevents overfitting, gives an unbiased estimate of model performance, and allows for parameter tuning.

Interview Questions

Dataset Types

- What is the difference between structured and unstructured data? Give examples.
- Why does unstructured data require special processing techniques?

Features and Labels

- Can you explain what features and labels mean in machine learning?
- Give a real-world example of features and labels in a dataset.

Feature Engineering

- What is feature engineering and why is it important?
- Describe some common feature engineering techniques.
- How can feature engineering help prevent overfitting?

Feature Scaling: Normalization & Standardization

- Why is feature scaling necessary in machine learning?

- What is the difference between normalization and standardization?
- How would you handle features with outliers?

Dimensionality Reduction
- What is dimensionality reduction, and why would you use it?
- Compare feature selection and feature extraction.
- Name and briefly explain two dimensionality reduction techniques.

Data Splitting
- Why do we split data into train, test, and validation sets?
- How would you decide the split ratio between these sets?
- What could go wrong if you tune hyperparameters with your test set?